MoleQuery

AI-Assisted Compound Research and Knowledge Management System

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1. Problem Statement

The Problem

Scientific compound information is distributed across research papers, documents and structured records. Researchers often need to manually search, read and organize this information.

Key Challenges

- Research information is scattered across multiple sources.
- Manual document searching is time-consuming.
- Compound–target relationships are difficult to track.
- Keyword search may miss semantically relevant information.

Impact

- Increased research time.
- Repeated reading of documents.
- Difficulty discovering relevant evidence.
- Poor organization of research knowledge.
- Difficulty converting documents into usable information.

Need

A centralized platform combining structured compound data, research documents and AI-assisted information retrieval.

2. Proposed Solution

What is MoleQuery?

The MoleQuery is a web-based system designed to centralize compound-related scientific information and make research knowledge easier to store, search, retrieve and understand.

Knowledge Management

- Manage compound records
- Manage biological targets
- Organize categories
- Store reference documents

Research Support

- Search compound information
- Filter research data
- Retrieve relevant documents
- Compare available information

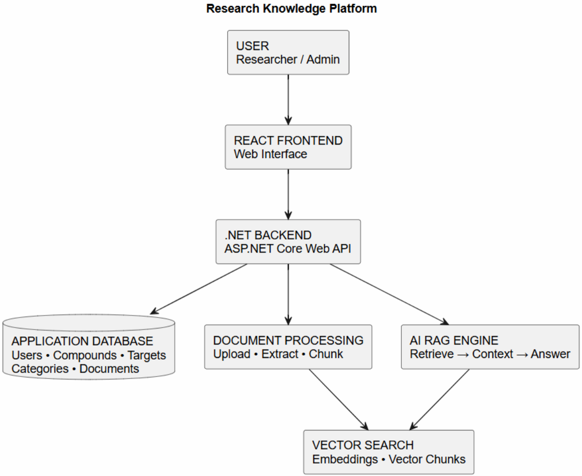
AI Assistance

- Semantic search
- Document retrieval
- RAG-based question answering
- Source-grounded responses

3. Functional Requirements

Module	Functional Requirements
Authentication	User registration, login and role-based access control.
Compound Management	Create, view, update and search compound records.
Target Management	Maintain biological targets associated with compounds.
Category Management	Classify compounds into relevant research categories.
Document Management	Upload, view and manage TXT, PDF and DOCX research documents.
Document Processing	Extract text, divide documents into chunks and prepare content for retrieval.
AI Research	Accept natural-language questions and retrieve relevant research context.
Search and Filtering	Search using compound name, synonym, formula, description, target and category.

4. System Architecture



5. AI Research System – RAG Workflow

Knowledge Preparation

1. Researcher uploads a document.
2. Text is extracted from the document.
3. Text is divided into smaller chunks.
4. Each chunk is converted into an embedding.
5. Embeddings are indexed for semantic retrieval.

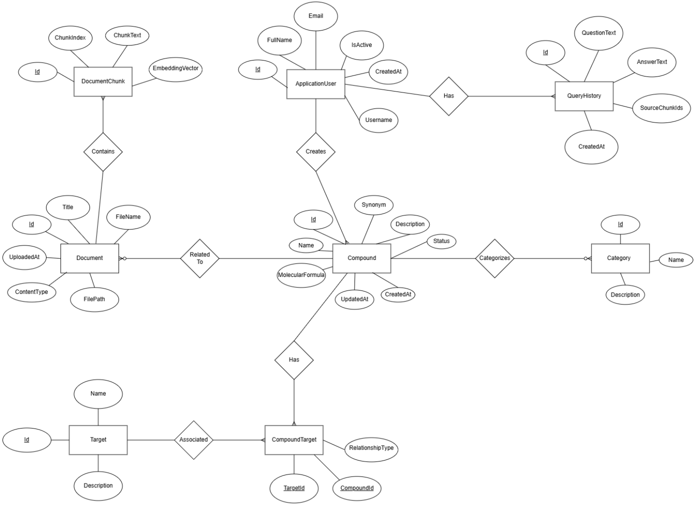
Question Answering

1. Researcher asks a natural-language question.
2. Query is converted into an embedding.
3. Similar document chunks are retrieved.
4. Retrieved context is provided to the AI model.
5. The model generates the final response.

RAG

Retrieval-Augmented Generation combines external research context with an AI language model to produce relevant and source-grounded answers.

6. Entity Relationship Diagram



7. Technology Stack

Frontend

- React.js + Vite
- Material UI
- Axios / API service layer

Backend

- ASP.NET Core Web API
- Entity Framework Core
- JWT Authentication

AI / Data

- Document Extraction
- Text chunking
- Embeddings
- Retrieval-Augmented Generation

8. Project Outcome and Future Scope

Project Outcome

- Centralized research knowledge base.
- Structured compound management.
- Faster information retrieval.
- AI-assisted research question answering.
- Semantic retrieval from research documents.
- Source-aware research responses.
- Role-based system access.

Future Scope

- Research-topic recommendation.
- Exportable research summaries.
- Advanced molecular comparison.
- Automated literature ingestion.
- Improved evidence ranking.
- Larger scientific knowledge bases.

Overall Contribution

The platform combines structured scientific data, document management and AI-powered semantic retrieval to make compound research more organized and efficient.

Thank You

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Questions?